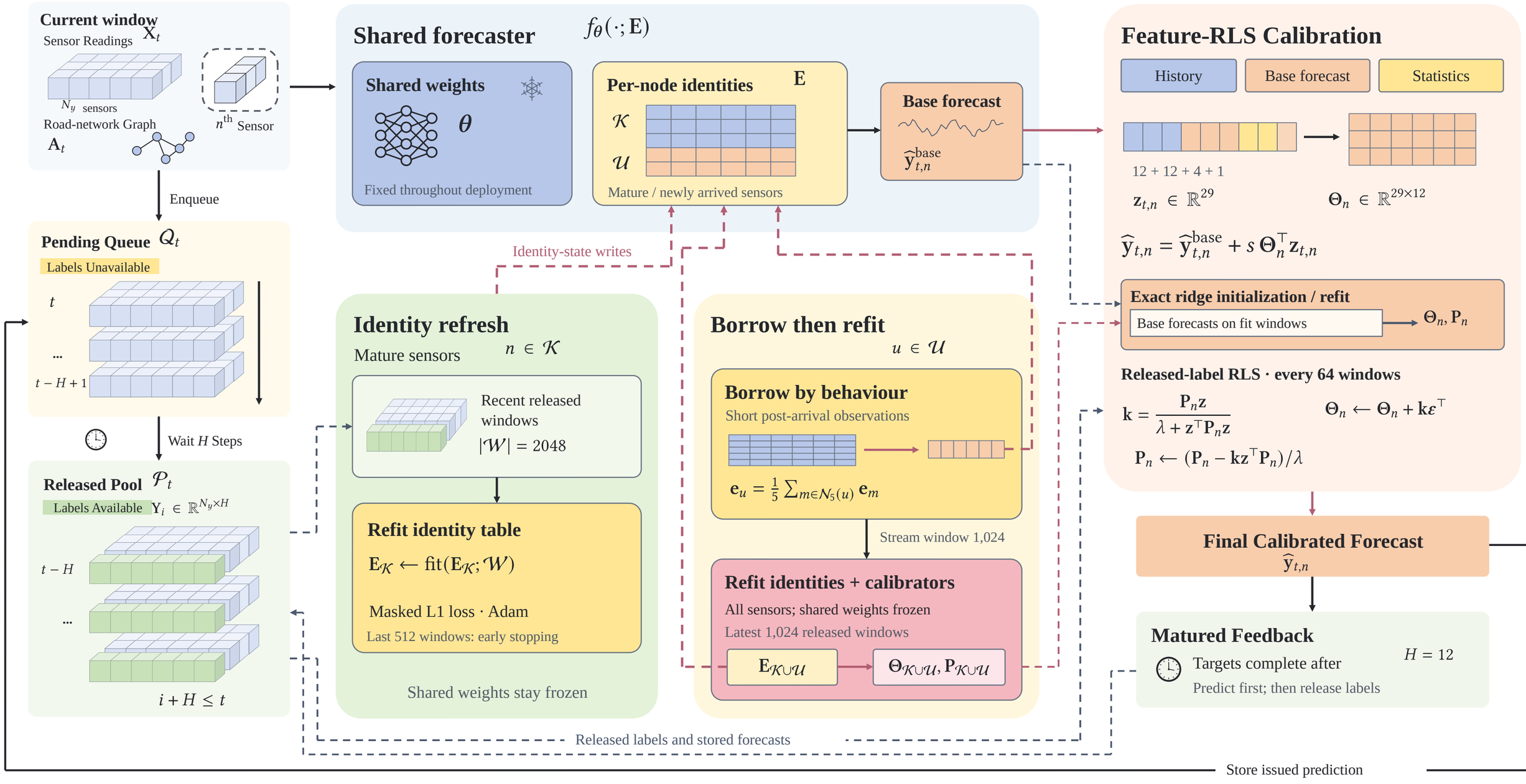

Current window
Sensor Readings X_t
N_y sensors
n^{th} Sensor
Road-network Graph
A_t
Enqueue
Pending Queue Q_t
Labels Unavailable
t
...
$t - H + 1$
Wait H Steps
Released Pool $\mathcal{P}_t$
Labels Available $Y_i \in \mathbb{R}^{N_y \times H}$
$t - H$
...
$i + H \le t$
Shared forecaster $f_\theta(\cdot; E)$
Shared weights
θ
Fixed throughout deployment
Per-node identities
E
$\mathcal{K}$
$\mathcal{U}$
Mature / newly arrived sensors
Base forecast
$\hat{y}^{base}_{t,n}$
Identity-state writes
Identity refresh
Mature sensors
$n \in \mathcal{K}$
Recent released windows
$|\mathcal{W}| = 2048$
Refit identity table
$E_{\mathcal{K}} \leftarrow \text{fit}(E_{\mathcal{K}}; \mathcal{W})$
Masked L1 loss · Adam
Last 512 windows: early stopping
Shared weights stay frozen
Borrow then refit
$u \in \mathcal{U}$
Borrow by behaviour
Short post-arrival observations
$e_u = \frac{1}{5} \sum_{m \in \mathcal{N}_5(u)} e_m$
Stream window 1,024
Refit identities + calibrators
All sensors; shared weights frozen
Latest 1,024 released windows
$E_{\mathcal{K} \cup \mathcal{U}}$
$\Theta_{\mathcal{K} \cup \mathcal{U}}, P_{\mathcal{K} \cup \mathcal{U}}$
Feature-RLS Calibration
History
Base forecast
Statistics
12 + 12 + 4 + 1
$z_{t,n} \in \mathbb{R}^{29}$
$\Theta_n \in \mathbb{R}^{29 \times 12}$
$\hat{y}_{t,n} = \hat{y}^{base}_{t,n} + s\, \Theta_n^\top z_{t,n}$
Exact ridge initialization / refit
Base forecasts on fit windows
Θ_n, P_n
Released-label RLS · every 64 windows
$k = \frac{P_n z}{\lambda + z^\top P_n z}$
$\Theta_n \leftarrow \Theta_n + k \varepsilon^\top$
$P_n \leftarrow (P_n - k z^\top P_n)/\lambda$
Final Calibrated Forecast
$\hat{y}_{t,n}$
Matured Feedback
Targets complete after
$H = 12$
Predict first; then release labels
Released labels and stored forecasts
Store issued prediction